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Interaction between preservatives and a monoclonal antibody in support of multidose formulation development

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ABSTRACT

Multidose formulations have patient-centric advantages over single-dose formats. A major challenge in developing multidose formulations is the prevention of microbial growth that can potentially be introduced during multiple drawings. The incorporation of antimicrobial preservatives (APs) is a common approach to inhibit this microbial growth. Selection of the right preservative while maintaining drug product stability is often challenging. We explored the effects of three APs, 1.1 % (w/v) benzyl alcohol, 0.62 % (w/v) phenol, and 0.42 % (w/v) m-cresol, on a model immunoglobulin G1 monoclonal antibody, termed the "NIST mAb." As measured by hydrogen exchange-mass spectrometry (HX-MS) and differential scanning calorimetry, conformational stability was decreased in the presence of APs. Specifically, flexibility (faster HX) was significantly increased in the C_H2 domain (HC 238–255) across all APs. The addition of phenol caused the greatest conformational destabilization, followed by m-cresol and benzyl alcohol. Storage stability studies conducted by subvisible particle (SVP) analysis at 40 °C over 4 weeks further revealed an increase in SVPs in the presence of phenol and m-cresol but not in the presence of benzyl alcohol. However, as monitored by size exclusion chromatography, there was neither a significant change in the monomeric content nor an accumulation of soluble aggregate in the presence of APs.

1. Introduction

Monoclonal antibodies (mAbs) capable of treating a wide variety of human diseases represent the fastest-growing segment in the biopharmaceutical industry (Kaplon and Reichert, 2021; Lagassé et al., 2017). The therapeutic mAb agents predominantly used in clinical trials are the immunoglobin G (IgG) class of mAbs, which is the largest category of immunoglobins (Ryman and Meibohm, 2017). During production and storage, mAbs can undergo biophysical and chemical modifications (Vlasak and Ionescu, 2008). A significant challenge in developing mAbs as therapeutic agents is the identification of solution conditions for long-term storage to preserve stability and solubility (Majumdar et al., 2013).

Multidose formulations are drug products packaged to permit the removal of multiple doses from a single container. The multidose format offer benefits over single-dose formats both from economic and patient-centric perspectives, as reviewed by Stroppel et al (Stroppel et al., 2023). In addition to a lower manufacturing cost, multidose formulations help to address potential issues related to dose preparation errors, drug product wastage, cost of goods for each container closure and delivery device. Finally, a multidose delivery device requires only a single dosage adjustment, which can improve patient compliance. These benefits have led to the preferred use of multidose formulations in the development of patient-centric biologics at a more affordable cost.

In the case of biologics, multidose formulations are liquids for injection. A major challenge in developing multidose formulations is the prevention of microbial growth that can potentially be introduced by penetration of the vial septum by a syringe needle during multiple drawings of the drug from a single vial. Even when a liquid multidose formulation is stored at below ambient temperature, ensuring sterility for days to weeks is difficult. The incorporation of antimicrobial preservatives (APs) in multidose formulations is a common approach to inhibit microbial growth. It can be challenging, however, to select the right preservative to support the multidose formulation and maintain adequate drug product stability while meeting regulatory requirements

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for a safe and efficacious drug product. In particular, the effect of preservatives on protein stability is a major concern.

Several studies have shown that APs seem to interact with proteins by stabilization of the unfolded states of potential aggregation sites that can lead to stability problems such as aggregation (Hutchings et al., 2013; Lim et al., 2009) either through direct AP-protein interactions or more globally by acting as a co-solvent. Zhang et al. showed that the incorporation of preservatives promotes aggregation by preferential binding to aggregation-prone regions for human granulocyte colony-stimulating factor (Zhang et al., 2015). Arora et al. reported a destabilizing effect of APs on the C_H2 domain of mAbs and correlated it with the hydrophobicity of preservatives (Arora et al., 2017). Interestingly, hydrogen exchange (HX)–mass spectrometry (MS) studies of IgG1 mAbs also found that the local flexibility of the C_H2 aggregation hotspot region is affected by other destabilizing factors (Majumdar et al., 2013; Manikwar et al., 2013; Thakkar et al., 2012).

In this work, we investigated the effect of three APs—benzyl alcohol, phenol, and m-cresol—on a model IgG1 mAb, the National Institute of Standards and Technology (NIST) mAb reference material (referred to here as the "NISTmAb"). The APs were selected on the basis of several factors. The AP must be compatible with the route of administration while being effective against various strains of fungi and bacteria. In addition, the preservatives must be active at the pH of the formulation. Additives such as polysorbates, inactive parabens, and phenolic preservatives also impose further restrictions. Additional concerns about the long-term antimicrobial efficacy of preservatives include poor aqueous solubility and concentration loss due to absorption to rubber stoppers. It is also important to consider acceptance of the preservatives by the target market regulatory agencies.

In this study, we used HX-MS and traditional formulation techniques to probe conformational changes in the presence and absence of APs. NISTmAb is a well-characterized protein that is ideal for incorporation as a model system. The results from this work will build a better understanding of biologic-preservative interactions to help improve the future development of multidose formulations.

2. Material and methods

2.1. Materials

NISTmAb (RM 8671), a humanized IgG1κ monoclonal antibody, was obtained from NIST (U.S. Department of Commerce, Gaithersburg, MD, USA) at a concentration of 10 mg/mL in 12.5 mM L-histidine, 12.5 mM L-histidine HCl at pH 6.0. All samples were prepared by diluting the NISTmAb stock solution in the selected preservatives under the following formulation conditions. Three samples were prepared by formulating the NISTmAb in effective preservative concentrations used in multidose parenteral formulations (Meyer et al., 2007) of 1.1 % (w/v) benzyl alcohol (10.1 mM), 0.62 % (w/v) phenol (6.6 mM), 0.42 % (w/v) m-cresol (3.9 mM) in 12.5 mM L-histidine, 12.5 mM L-histidine HCl at pH 6.0.

2.2. Bradford assay

To determine whether the mAb precipitated in the presence of APs, NISTmAb concentrations were determined by using a microplate based Bradford assay. Protein samples at a concentration of 1.8 mg mL^{-1} with and without APs were centrifuged at 10,000 × g for 10 min at room temperature. From the supernatant of each sample, 4 μL was pipetted into microplate wells. To each well, 300μL of Coomassie Plus Reagent (Thermo Fisher Scientific, Waltham, MA, USA) was added and mixed well. The plate was incubated for 10 min at room temperature and the absorbance was then measured in triplicate at a wavelength of 595 nm, using a plate reader (VersaMax; Molecular Devices, San Jose, CA, USA). The average absorbance measurement for the blank replicates was subtracted from that of the individual sample replicates.

Mean absorbance values from the Bradford assay for 1.8 mg/mL NISTmAb in the presence and absence of preservatives are shown in Table S1. No appreciable difference was seen in the absorbance values of different formulations, indicating that the APs did not cause a significant change in protein concentration for the samples containing preservatives with respect to the control. Therefore, the results from the Bradford assay indicate that there was no significant precipitation of the protein due to the presence of preservatives at the low concentrations used in HX-MS studies.

2.3. Accelerated stability study

Samples of NISTmAb in the presence of APs at 1.8 mg/mL, along with a control sample without preservative, were filtered with a 0.22-μm pore Stericup filter (Merck Millipore, Billerica, MA, USA). Under aseptic conditions in a biosafety cabinet, 1-mL aliquots were placed into sterile 2R glass vials, which were sealed with rubber stoppers. Prepared samples were then incubated under accelerated storage stability conditions at a temperature of 40 °C. One vial of each formulation was retained for characterization at time zero. The remaining samples were retrieved from the stability chamber for analysis after 1, 2, 3 and 4 weeks of incubation in the dark at 40 °C.

2.4. Subvisible particle count by microflow imaging

An MFI 5000 particle imager (ProteinSimple, San Jose, CA) was used to monitor the subvisible particle (SVP) count of the samples based on direct measurement on the neat samples The instrument was cleared between samples by running water and 5 % Liquinox (MilliporeSigma, Burlington, VT) through the sample cell.

2.5. Size exclusion chromatography

The relative percentage aggregate, monomer and fragment of samples at each time point was determined by size exclusion chromatography (SEC), using a high-performance liquid chromatography system with a TSKGel G3000 column (Agilent, Santa Clara, CA, USA). A molecular-weight standard (Bio-Rad Laboratories, Hercules, CA, USA) was used to calibrate the column, and the IgG1 mAb NIP228 (AstraZeneca) was used as a reference standard. All samples and standards were centrifuged at 13,000 × g for 1 min in 0.45-μm centrifugal filter units (Merck Millipore, Billerica, MA, USA) to remove insoluble aggregates prior to injection. A mobile phase containing 0.1 M sodium phosphate dibasic anhydrous and 0.1 M sodium sulfate at a flow rate of 1 mL/min was used to separate mAb species. An injection volume of 100 μL was used to ensure that 180 μg of protein was loaded onto the column per injection. The length of each run was 25 min, and the automatic integration of peaks for fragment, monomer, and aggregate was performed using ChemStation (Agilent).

2.6. Differential scanning calorimetry

Differential scanning calorimetry (DSC) thermograms were obtained over a temperature range of 10–90 °C at a scan rate of 60 °C h^{-1}, using a VP-Capillary DSC with an autosampler (MicroCal, Northampton, MA, USA). All samples were measured in triplicate at a NISTmAb concentration of 0.5 mg/mL in 12.5 mM L-histidine, 12.5 mM L-histidine HCl at pH 6.0. Thermograms of NISTmAb in the absence (control) and presence of preservatives 1.1 % (w/v) benzyl alcohol, 0.62 % (w/v) phenol, and 0.42 % (w/v) m-cresol were compared with the reference thermograms of buffer alone containing no protein. Data were analyzed with the Origin 7.0 software package (OriginLab, Northampton, MA, USA). T_{onset} was determined by identifying the point where the heat capacity (C_p) value for the first thermal transition reached 500 cal mol^{-1} °C^{-1}. The transition temperatures were determined using a standard curve fitting method in the Origin software.

2.7. HX-MS

Stock NISTmAb solution was diluted to a concentration of 1.8 mg mL^{-1} in 12.5 mM L-histidine, 12.5 mM L-histidine HCl at pH 6.0. An internal exchange reporter compound, C-5–substituted 1,3-dimethyl benzimidazolium derivative (TM-65) was received as a gift from Miklos Guttman (University of Washington, Seattle, WA, USA) (Murphree et al., 2020). The TM-65 internal exchange reporter compound was added to the protein solution at a final concentration of 6 mM.

The deuterated labeling buffer for the HX-MS study was prepared with either no preservatives (control) or 1.1 % (w/v) benzyl alcohol, 0.62 % (w/v) phenol, 0.42 % (w/v) m-cresol, and TM-65 in 12.5 mM L-histidine, 12.5 mM L-histidine HCl in D_2O at pD 6.0 (pH reading incorporating the isotope effect) (Glasoe and Long, 1960). HX-MS measurements were carried out as previously described (Rincon Pabon et al., 2021), with the following adjustments. Eight microliters of protein solution was diluted 10-fold with labeling buffer and incubated at 25 °C for six exchange times: 30, 100, 500, 2500, 12500, and 62500 s, prepared in triplicate. At each time point, hydrogen exchange was quenched at 0 °C by 1:1 dilution with pH 2.50 solution containing 0.5 M tris(2-carboxyethyl)phosphine–3.0 M guanidine hydrochloride–200 mM glycine. Quenched samples were held for 500 s before analysis by liquid chromatography (LC)–MS. Deuterium labeling reactions were carried out using a robotic liquid handler (HDX PAL; LEAP Technologies, Carrboro, NC, USA). Experimental data for mass shifts were analyzed and manually reviewed using HDExaminer, version 3.2 (Sierra Analytics, Modesto, CA, USA). To evaluate statistically significant differences between the control NISTmAb sample (no preservatives) and samples with preservatives, a hybrid significance testing approach was used (Hageman and Weis, 2019). The individual mean peptide mass difference ΔD was determined as equation (1):

$$\Delta D = m_{\mathrm{AP}} - m_{\mathrm{control}} \tag{1}$$

where m_{AP} and m_{control} are mean peptide centroid masses from replicate labeling reactions for the same peptide from NISTmAb in the presence of AP and control, respectively. Statistically significant differences in HX were mapped to a homology model of NISTmAb (Protein Data Bank [PDB] identifier, 1HZH).

Peptic peptide mapping of NISTmAb was carried out by data-dependent and targeted LC–collision-induced dissociation–tandem MS on an Agilent 6530 Q-TOF mass spectrometer. A 12-min linear gradient of aqueous 0.1 % formic acid in acetonitrile increasing from 1 % to 35 % acetonitrile was used, followed by a ramp to 95 % acetonitrile for cleanup. To minimize carryover, the trap and column were cleaned after the elution, using an independent secondary gradient. Agilent BioConfirm B.07.00 was used to match sequences of both samples with tandem mass spectra after applying relevant modifications to map the peptic peptides.

2.8. Chemical exchange reporter compounds

HX kinetic plots for the internal exchange reporter compound in the presence and absence of preservatives are shown in Fig. S2. Because this compound lacks secondary structure, any variation in deuterium uptake suggests a change in exchange reaction conditions that alters the chemical exchange rate (Murphree et al., 2020). The uptake plots show no appreciable difference in deuterium uptake for the internal exchange reporter in the presence and absence of the three APs used in the study. This observation indicates that the chemical HX rates were not altered by the formulation conditions used in the experiment, and thus, all observed differences in HX caused by preservatives can be attributed to changes in conformation and dynamics as opposed to differences in chemical exchange rates caused by the preservatives.

3. Results

Physical appearance (color, clarity, and visible particles) of the NISTmAb in the presence of APs over 1 month period of incubation at 40 °C was analyzed by visual inspection. No immediate change in sample turbidity from the NISTmAb control was observed before and after filtration following the addition of phenol, m-cresol, or benzyl alcohol. All samples remained clear for the incubation period of 1 month at 40 °C when formulated with m-cresol, phenol, or benzyl alcohol. Samples containing benzyl alcohol, m-cresol, and the control contained fewer than 10 visible particles. The visible particles in samples containing phenol were too numerous to count and were equivalent to those of an in-house standard containing 520 visible particles per milliliter. NISTmAb in the presence of benzyl alcohol, along with the control, turned from colorless to slightly yellow after 1 month of incubation at 40 °C, whereas samples containing m-cresol or phenol turned from colorless to sightly brown (results not shown).

Microflow imaging was used to determine the number of SVPs in samples of NISTmAb in the presence of APs. The SVP count is an indication of the number of insoluble aggregates. Measurable increases in SVPs occurred in all four formulations, but the extent of particle formation was substantially different between formulations (Fig. 1). Both phenol and, to a lesser extent, m-cresol induced substantial formation of SVPs, whereas samples formulated with benzyl alcohol and the control formulation had much lower levels of particle formation over 4 weeks.

SEC was used to monitor changes in fractional composition of monomer, soluble aggregate, and fragment content of NISTmAb samples over 4 weeks at 40 °C (Fig. 2). (See Figure S1 in the Supporting Information.) Over 4 weeks, there was only an approximately 1 % decrease in monomeric mAb, from approximately 98 % to 97 %, and a corresponding 1 % increase in fragment over 4 weeks. A small fraction (less than 1 %) of soluble aggregate was detected but decreased to undetectable levels over 4 weeks. Overall, based on the SEC measurements of soluble species, there were no substantial differences in the composition of soluble forms of NISTmAb between samples formulated in the presence or absence of the APs, and therefore little to no effect of any of these preservatives observed relative to the control formulation over 4 weeks at 40 °C. Taken together, the microflow imaging and SEC data indicate that the primary route of degradation was in the formation of insoluble SVPs detected through microflow imaging.

The thermal stability of the protein in the presence of APs was studied by DSC. DSC analysis of NISTmAb in the presence of the three APs indicated three distinct thermal unfolding events. The onset temperature (the temperature at which protein unfolding starts) and transition temperatures T_m1, T_m2, and T_m3 from triplicate runs are listed in Table 1. These three separate transition temperatures are associated with the unfolding of the C_H2, F_{ab}, and C_H3 domains, although the individual transitions cannot be assigned to specific individual unfoldings (Vermeer and Norde, 2000; Vermeer et al., 2000). In contrast to the results from accelerated stability, measured by SEC, the APs did have discernable effects on thermal stability. A decrease in unfolding temperatures (T_{onset}, T_m) was observed with all the NISTmAb samples in the presence of APs compared with the control sample, indicating a loss of conformational stability caused by the preservatives. As indicated by T_m and T_{onset} values, all three preservatives decreased the thermal stability of the NISTmAb. Notable differences were observed in the effects of the three preservatives on the T_m1 and T_m3 peaks. With both of these transitions, benzyl alcohol was slightly more destabilizing than the other two preservatives. No substantial difference was observed between the effects of m-cresol and phenol on NISTmAb thermal stability.

We used differential HX-MS analysis to understand the effect of APs on the local backbone flexibility of NISTmAb. Separate experiments were completed with NISTmAb in the presence and absence of three APs. Representative deuterium uptake curves for six peptides from across different domains of NISTmAb in the presence and absence of the three APs are shown in Fig. 3.

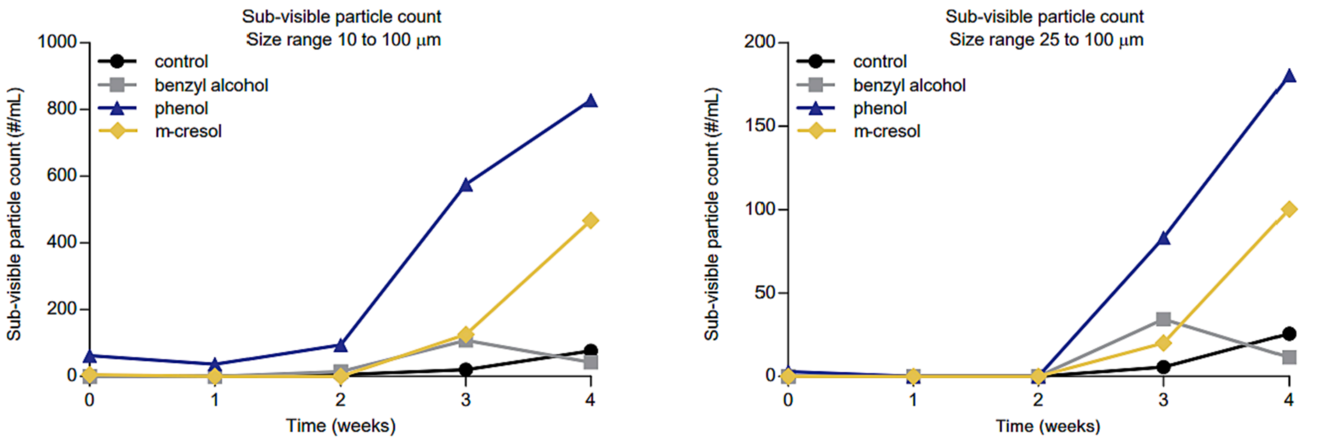

Fig. 1. Sub-visible particle formation, as measured by microflow imaging, over 4 weeks at 40 °C. The results are based on measurements of individual, independent samples at each time point. Particle counts in microflow imaging are based the detection and sizing of many individual particles in each independent sample.

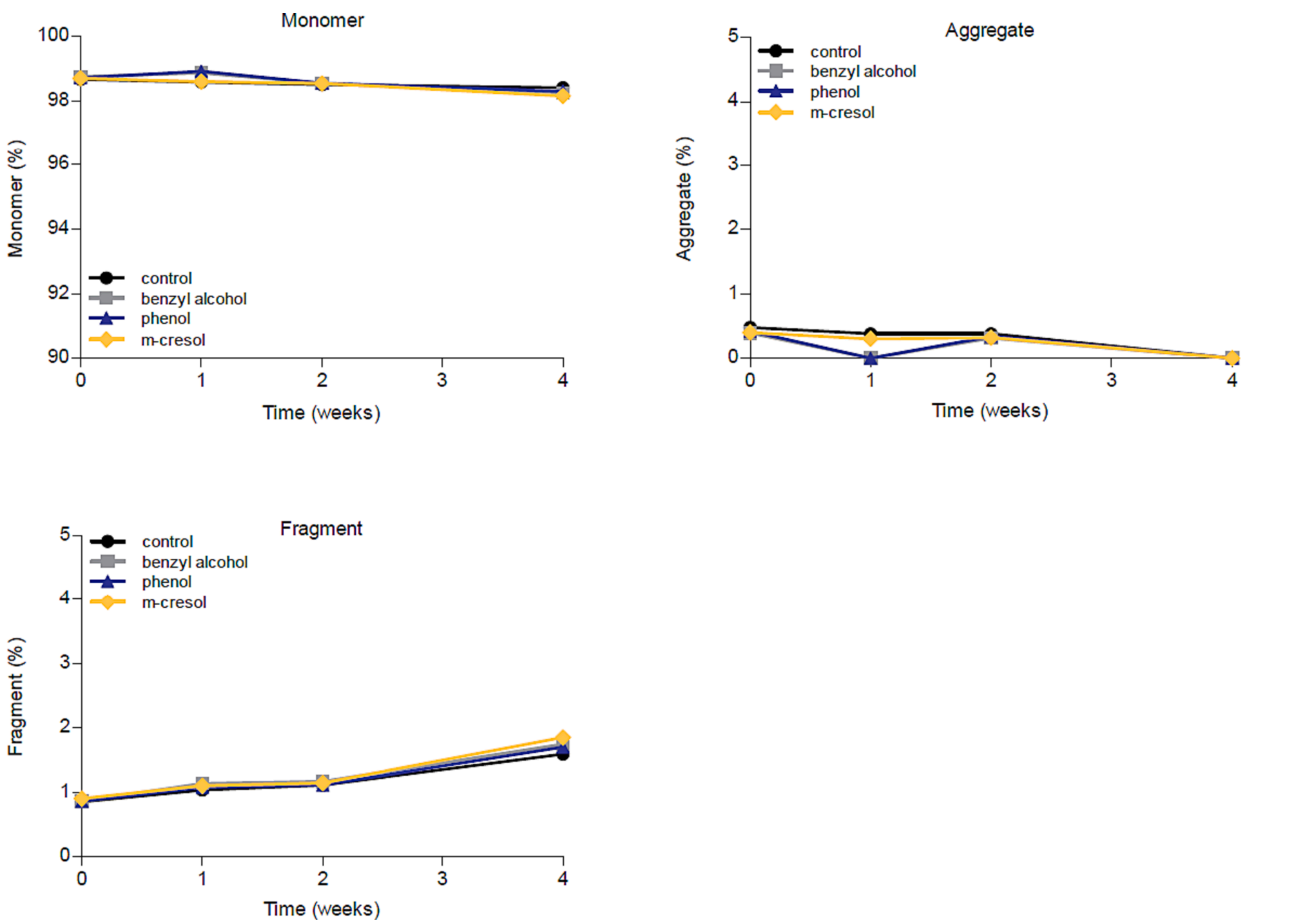

Fig. 2. Fractional speciation of soluble forms of NIST (monomer, soluble aggregate, and fragment) mAb under accelerated stability conditions as determined by SEC. Each data point represents duplicate technical replicate measurements of a single sample.

The presence of APs caused both slower and faster hydrogen exchange in different regions of the NISTmAb. A positive value for the mean peptide mass difference ΔD in equation (1) indicates an increase in flexibility (i.e., "deprotection") for the peptide relative to the control, and a negative value represents a decrease in flexibility (i.e., "protection"). Significant differences in deuterium uptake were identified by hybrid significance testing (Hageman and Weis, 2019) at $\alpha = 0.01$ (Fig. S3).

Although the APs did not significantly alter HX kinetics in the majority (>90 %) of the peptides, some regions did exhibit significant changes in HX kinetics. Fig. 4 shows the HX-MS results mapped onto a homology model of NISTmAb (based on PDB 1HZH), and Table 2 summarizes the results with respect to patterns that were common to or distinct from the APs. The presence of benzyl alcohol caused a subtle but statistically significant decrease in HX kinetics in the heavy-chain variable (V_H) and C_H1 domains, residues 118–145 and 149–177 (blue

Table 1
Thermal stability of the NIST mAb at 0.5 mg/mL.

Preservative	Temperature (°C)			
	T_{onset}	T_m1	T_m2	T_m3
Control	60.3 ± 0.8	68.3 ± 0.1	82.6 ± 0.1	93.5 ± 0.1
benzyl alcohol (1.1 %, w/v, 10.1 mM)	55.1 ± 0.2	62.8 ± 0.1	78.5 ± 0.1	90.4 ± 0.1
m-cresol (0.42 % w/v, 3.9 mM)	55.8 ± 0.5	63.5 ± 0.1	78.7 ± 0.1	91.2 ± 0.1
phenol (0.62 % w/v, 6.1 mM)	56 ± 2	63.2 ± 0.2	78.3 ± 0.1	91.0 ± 0.1

The uncertainty represents the standard deviation based on three technical replicates.

regions in Fig. 4). In contrast, there was a significant increase in the rate of HX in the C_H2 domain, residues 238–255 (yellow region in Fig. 4). In addition, benzyl alcohol caused widespread, but not statistically significant, slowing of HX throughout the mAb, as is evident from a negative bias in the distribution of ΔD values (Fig. S3).

Similarly to benzyl alcohol, m-cresol induced an obvious increase in flexibility in the C_H2 domain involving residues 238–255. The magnitude of the effect in this region was stronger than for benzyl alcohol (Fig. 3). In addition, the change induced by m-cresol extended into the 264–280 region of the C_H2 domain, suggesting a stronger interaction of the mAb with m-cresol than with benzyl alcohol. Indeed, the effect of phenol was even stronger (see following paragraph). For all preservatives, no significant difference was observed in deuterium uptake in the 256–263 region of the C_H2 domain. In the constant domain of F_{ab}, the 135–171 region also became deprotected in the presence of m-cresol, an effect that was not detected in the mAb in the presence of the other two preservatives. Finally, unlike the other two preservatives, m-cresol did not induce any protection.

Overall, the effects induced by phenol on NISTmAb flexibility in the Fc region were more extensive and of greater magnitude than those caused by benzyl alcohol and m-cresol. With phenol, the affected Fc regions were the 238–280 and 304–351 regions of the C_H2 domain and the 372–382 and 414–449 regions of the C_H3 domain. Interestingly, the 118–145 region in the F_{ab} V_H domain became weakly protected, similar to benzyl alcohol, an effect that was not observed for NISTmAb formulated with m-cresol. However, unlike the effect of benzyl alcohol, which also caused protection in the C_H1, 149–177 region, phenol did not induce significant protection in the C_H1 domain. Overall, a complex pattern of AP effects on the HX of the NISTmAb was apparent with both protection and deprotection. All three APs induced faster HX in the C_H2 domain, albeit with different magnitudes and to different extents. The effect in the 238–255 region was the most pronounced and common across all three APs. Except for benzyl alcohol, this effect also extended into the adjacent 264–280 region.

4. Discussion

The purpose of this work was to better understand how the addition of preservatives to an IgG1 mAb formulation alters the molecular structure of the mAb. HX-MS was employed as the main technique to measure changes in the structure and flexibility of the molecule in solution at a resolution that enabled the desired molecular-scale details. We investigated the effects of three APs, benzyl alcohol, m-cresol and phenol, on this model IgG1 mAb, the NISTmAb. Since most of the effects of these preservatives on HX kinetics were detected in the constant domains, it is likely that the observed effects represent a common pattern of altered backbone flexibility in IgG1 mAbs.

Benzyl alcohol was the most thermally destabilizing but induced the least insoluble particle formation. Phenol was the least thermally destabilizing but induced the fastest formation of insoluble particles. Thus, DSC was not predictive of the observed particle formation under thermal stress. Based on SEC data obtained on thermally stressed samples, there was no discernable difference in physical stability, despite clear differences in thermal stability; therefore, DSC results did not correlate with those of the accelerated stability study.

Goyal et al. also observed stabilization of a model protein (lysozyme) under thermal stress with the addition of benzyl alcohol but a concomitant reduction in the DSC-measured melting temperature (Goyal et al., 2009). They postulated that this might be due to benzyl alcohol stabilizing a partially unfolded intermediate that prevents progression to a fully unfolded aggregate formation. According to the law of mass action, preferential binding of benzyl alcohol to the partially unfolded state will lead to destabilization of protein (Arakawa and Tsumoto, 2003), resulting in a reduced melting temperature as measured by DSC. It should be noted that in the Goyal et al. study, lysozyme activity was not affected by stabilization of this potentially partial unfolded state by benzyl alcohol (Goyal et al., 2009).

HX effects in the C_H2 domain did correlate with increased formation of insoluble aggregates (benzyl alcohol < m-cresol < phenol), suggesting that the C_H2 domain may play an important role in aggregation for IgG1 mAbs. Other HX patterns, for example V_H domain protection, were not correlated with SVP formation.

Arora et al. probed the effect of four similar APs on an IgG1 mAb, using a fixed molar concentration of the preservatives (Arora et al., 2017) rather than keeping the preservative concentrations at pharmaceutically relevant values. However, from a pharmaceutical perspective, the relevance of using equal molar concentrations for APs may be less useful. In the work presented here, we examined the behavior of APs on another IgG1 mAb at distinct, pharmaceutically relevant concentrations of the preservatives: 1.1 % (w/v) benzyl alcohol, 0.62 % (w/v) phenol, and 0.42 % (w/v) m-cresol. In comparison to earlier work, some of the differences observed in this study can be related to the differences in the concentrations used. Thus, our work sheds light on understanding the effects of preservatives at relevant concentrations that are used in multidose formulations.

In contrast with Arora et al.(Arora et al., 2017), we did not observe a correlation between conformational stability and aggregation propensity following the ranking of hydrophobicity (logP values), as was previously found by those investigators. Instead, the greatest conformational destabilization in the C_H2 region was caused by the presence of phenol, followed by m-cresol and benzyl alcohol. Increased flexibility in the 238–255 region of the C_H2 domain, including the addition of chaotropes, oxidation of distal methionines, deglycosylation, and changes in pH, has been frequently identified in HX-MS investigations of a variety of studies of IgG1 mAbs (Arora et al., 2015; Houde et al., 2009; Hu et al., 2020a; Hu et al., 2020b; Iacob et al., 2013; Majumdar et al., 2015; Mo et al., 2016; More et al., 2018; Toth et al., 2018; Zhang et al., 2018). In other work, this region has been identified as a potential aggregation hotspot in IgG1 mAbs (Chennamsetty et al., 2009). Notably, the T_m1 transition corresponding to the C_H2 domain was the most prominent effect detected by DSC, and deprotection in C_H2 was the most prominent effect detected; however, the change in T_m1 seemed to be inversely correlated with the magnitude of the HX effect in the 238–255 region (Table 2, Fig. 3).

This destabilization of the NISTmAb seen by HX was also reflected in the assessments of physical stability. Stress (40 °C) storage conditions are used in mAb development to accelerate the degradation pathways typically observed under real-time storage conditions and can be used to predict the degradation rates by application of an Arrhenius model extrapolation. After 1 month at 40 °C, the visible particle count in the presence of phenol was noticeably higher than in samples containing m-cresol and benzyl alcohol, suggesting that phenol induced insoluble aggregate formation. A change from colorless to slightly brown was observed in samples with phenol and m-cresol, which is typically seen with oxidization of phenolic compounds.

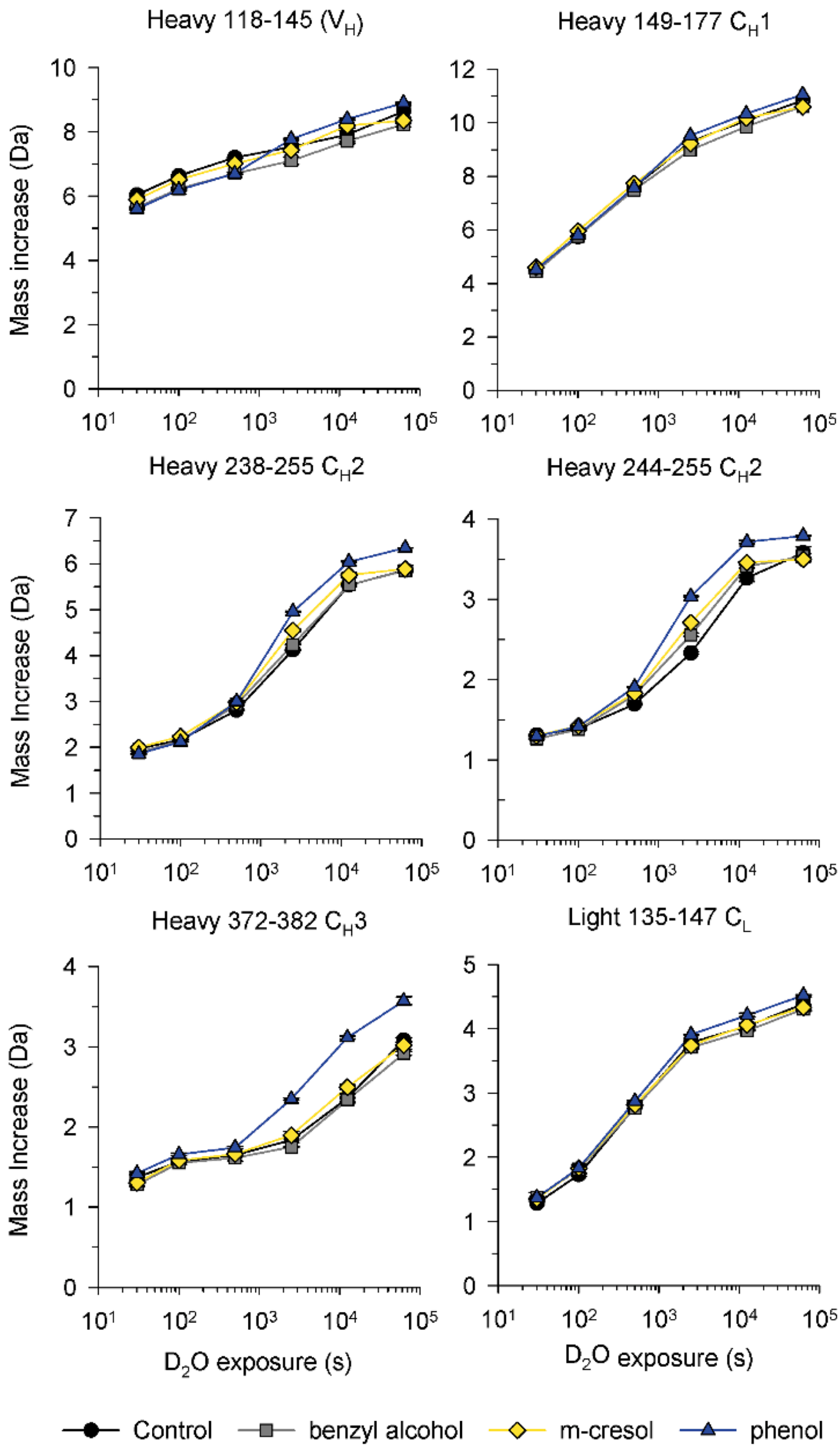

Fig. 3. Uptake plots for HX by the NISTmAb in different formulations from representative regions of the light and heavy chains. Measurements are based on triplicate independent exchange reactions. Error bars, representing one standard deviation, are smaller than the symbols.

5. Conclusions

Among the preservatives tested, benzyl alcohol has been shown here to be the least detrimental on formulation stability, potentially even improving the structural stability of the NISTmAb to a small extent. In this work HX data was used to determine the impact of different preservatives/excipients on protein structure. In future, HX could be used to gain further structural insights into excipient-protein interactions to facilitate screening of the stabilizing potential of new formulation excipients on the next generation of biopharmaceutical therapeutics.

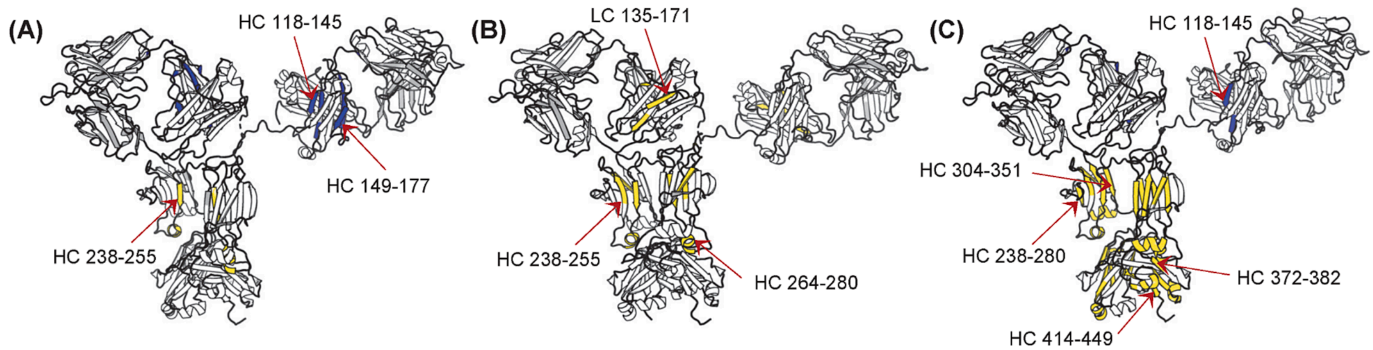

Fig. 4. Significant HX effects mapped onto a homology model of the NISTmAb based on the IgG1 structure 1HZH with addition of benzyl alcohol (A), m-cresol (B) or phenol (C). Significantly faster hydrogen exchange is displayed in yellow, significantly slower hydrogen exchange is displayed in blue. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 2
Locations of statistically significant changes in HX induced by preservatives.

Preservative	Deprotected regions		Protected regions	
	Common	Distinct	Common	Distinct
benzyl alcohol (1.1 %, w/v)	HC 238–255 (C_H2)	None	HC 118–145 (V_H)	HC 149–177 (C_H1)
m-cresol (0.42 % w/v)	HC 238–255 (C_H2), HC 264–280 (C_H2)	LC 135–171 (C_L)	None	None
phenol (0.62 % w/v)	HC 238–280 (C_H2)	HC 304–351 (C_H3), HC 372–382 (C_H3), HC 414–449 (C_H3)	HC 118–145 (V_H)	None

CRediT authorship contribution statement

Sachini P. Karunaratne: Investigation, Data curation, Writing – original draft. **Madeleine C. Jolliffe:** Investigation, Data curation, Writing – original draft. **Isabelle Trayton:** Writing – review & editing. **Ramesh Kumar Shanmugam:** Conceptualization, Methodology. **Nicholas J. Darton:** Conceptualization, Methodology, Writing – review & editing. **David D. Weis:** Conceptualization, Methodology, Writing – review & editing.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: [Madeleine C. Jolliffe reports financial support was provided by AstraZeneca PLC. Isabelle Trayton reports financial support was provided by AstraZeneca PLC. Ramesh Kumar Shanmugam reports financial support was provided by AstraZeneca PLC. Nicholas J. Darton reports financial support was provided by AstraZeneca PLC. David Weis reports financial support was provided by Bristol Myers Squibb].

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijpharm.2023.123600.

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